Skony — Track 2A OpenParlData: Technical Report

Repository: <https://■/■github.■com/■ahmarkhhan/■sfedly> (built from the HackApertus project template).■ Build and completion evidence: [BUILD_■STATUS](docs/■BUILD_■STATUS.■md).■ Shared affair records and evaluation remain planned.■ Response caching, batch checkpoints, saved-■result import and gold-■export safeguards are implemented.■ Docker packaging and a synthetic practice replay exist; actual container execution is still unverified.■ [Pipeline evidence](docs/■PIPELINE_■VERIFICATION.■md).

Status: extraction and local review implementation, 16 October 2026.■ One real-■document pilot is complete; dataset-■level extraction accuracy has not been measured.■ Measured run evidence: [RESULTS](docs/■RESULTS.■md).

Judging criteria and where they are addressed

Criterion (official brief wording)	Where it is addressed	State
Insight: a well-■founded conclusion on common-■schema feasibility; partial or negative counts	[Honest limits](#honest-■limits); [RESULTS](docs/■RESULTS.■md#honest-■limits)	Partial: a common raw-■field representation proved feasible for the eight-■field slice tested; the unresolved seam is question/■answer pairing and role assignment across bodies
Breadth: demonstration across document types, languages and levels of government	[RESULTS §Measured runs](docs/■RESULTS.■md#measured-■runs); corpus integrity table in [PIPELINE_■VERIFICATION](docs/■PIPELINE_■VERIFICATION.■md)	Narrow but real: DE + FR, two bodies (municipal Nidau, cantonal Genève), born-■digital plus one scanned/■OCR-■layer document; a 44-■PDF hash-■verified corpus awaits review
Traceability and faithfulness: no invented values; span-■level provenance; flag rather than guess	[Implemented first slice](#implemented-■first-■slice); `src/■skony/■verify.■py` (abstains on unlocated, fabricated or ambiguous evidence); strict-■validation rejection tests in `tests/■test_■evidence.■py`	Implemented: every non-■null field carries document ID, page, original quote and character offsets; unverifiable values abstain
Built with Apertus	`src/■prompts/■extract.■txt`; [RESULTS](docs/■RESULTS.■md) (recorded `swiss-■ai/■Apertus-■v1.■5-■70B` runs with token and latency observations)	Yes, at CSCS; development-■time only, not redistributed
Reusable outputs: public repository with converter, prompts, validator and README samples	[README](README.■md) quick-■start; [RESULTS §Reproduce](docs/■RESULTS.■md#reproduce-■public-■source-■no-■credentials-■no-■docker); hash-■checked synthetic fixture in `data/■sample/■`; 131-■test offline suite	Yes, without credentials or network: replay reproduces recorded results byte-■identically
Judge execution: `make run` in Docker	root and track `Makefile`, `Dockerfile`, `compose.■yaml`, `scripts/■check_■docker.■py`, CI workflow `.■github/■workflows/■docker-■verify.■yml`	Packaged, unverified on the development host (no Docker/■WSL there); the clean public-■source start passes outside a container

Problem and contribution

Investigate which parliamentary affair fields can share a common representation across bodies and languages.■ Preserve original wording, explicitly abstain when evidence cannot be located, and record where the shared schema cannot represent the source faithfully.■ Our partial conclusion: a common raw-■field representation is feasible for the eight-■field slice tested — fields were located with original wording and spans across one DE municipal and one FR cantonal document — while question/■answer pairing and role assignment across bodies remain the unresolved seam.

Implemented first slice

PDF → page text and digest → Apertus schema-■constrained prompt → strict JSON validation → evidence lookup → review-■required JSON.■ The output has eight raw fields.■ Normalisation and controlled vocabularies are planned, rather than inferred and silently treated as source text.

Each non-■null result includes a document ID, page number, original quote and character offsets into the unchanged extracted page text.■ `source_■located` means a string was found; it does not establish correct field assignment or faithfulness of question/■answer boundaries.■ All such results require review.

Validation so far

131 engineering tests passed on local Python 3.■11 ([RESULTS](docs/■RESULTS.■md)).■ The client was exercised with mocked HTTP transports and the real CSCS service.■ Tests cover fabricated quotes/■values, wrong page attribution, repeated quotes, missing evidence, textless/■oversize input, malformed schemas, truncated/■mismatched endpoint responses, line-■break span recovery and obviously corrupt submitter OCR.■ Review tests also cover persistence, stale revisions, fabricated correction rejection, explicit absence, visual transcription, audit snapshots, model reruns preserving human decisions, reviewed-■only export and local route/■secret isolation.

New execution checks cover replay blocking live calls, synthetic-■fixture hashes, restart persistence and mapped localhost Host/■Origin checks.■ A clean public-■source copy starts without private data/■keys and serves the PDF/■image and reviewed export.■ The bundled practice data are authored examples, not model output or evaluation gold.

The batch/■cache checks cover full effective-■request identity, corrupt-■cache rejection, schema-■invalid response replay, actual call accounting, preserved failed downloads, unchanged manifests, interrupted resume and idempotent workbench import.■ Gold-■export checks exclude synthetic/■QA records and prevent stale labels from a previous export directory entering a new manifest.

An approved real-■data engineering proof processed one DE Nidau PDF and one FR Genève PDF using Apertus 1.■5 70B at CSCS.■ The first run made two calls, reporting 3,953 total tokens and 13.■594/■9.■406 seconds.■ Offline replay, with no credential and an HTTP-■rejecting transport, used two cached responses and produced identical verified results with zero model calls.■ One failed-■fetch row remained in both ledgers.■ This demonstrates pipeline reuse, not semantic accuracy or multilingual benchmark coverage.■ Both results were imported without altering existing review decisions.■ The store has 12 decisions across four documents, one mechanically complete; it is not yet a frozen, competence-■checked evaluation gold set.

On one scanned Nidau postulate with an existing OCR layer, the 8B pilot first copied the schema and then returned inconsistent null/■evidence combinations.■ Both were rejected.■ The 70B pilot returned valid fields.■ Replaying that recorded response with verifier `evidence-■v0.■2` located the title, type, printed date and request; it abstained on the garbled submitter and absent answer/■decision fields.■ The 70B call took 7.■234 seconds and used 1,725 tokens.■ These are single-■call observations, not a benchmark or a gold-■set result.

Planned evaluation

Use separate development and held-■out human-■reviewed documents spanning German and French and multiple parliamentary bodies.■ Compare a fair simple baseline, raw Apertus extraction and extraction with evidence checks.■ Report field precision/■recall, abstention/■coverage, source-■location accuracy, question/■answer pairing, latency and actual token usage.■ Scans and minority layouts must remain visible in the report rather than being filtered out of the denominator.

Honest limits

Each bullet states a limitation, its consequence and where the evidence is.■ This section, not the feature list, is the insight contribution.

- **Two documents are not coverage.**■ Model responses exist for exactly two

real PDFs, both team-■selected.■ Nothing measures accuracy on unseen documents; breadth above is corpus breadth, not measured extraction breadth.■ Evidence: [RESULTS](docs/■RESULTS.■md).

- **In-■sample tuning.**■ The prompt (`raw-■v0.■2`) and verifier (`evidence-■v0.■2`)

were shaped on the same pilot documents that demonstrate them.■ The replay results are self-■consistent by construction, not evidence of generalisation.■ Evidence: [PIPELINE_■VERIFICATION](docs/■PIPELINE_■VERIFICATION.■md).

- **No gold, no rates.**■ Twelve human field decisions exist but are not

competence-■checked, and one reviewed source language is unrecorded.■ Precision, recall and abstention rates would be fabricated today, so none are presented.■ Evidence: [RESULTS §Review and gold-■set

state](docs/■RESULTS.■md#review-■and-■gold-■set-■state).

• **Located is not correct.■** `source_■located` proves a string match, not

semantic assignment: a value can occupy the right span and the wrong field.■ Every located field therefore requires review; question/■answer pairing is unvalidated.■ Evidence: [Implemented first slice](#implemented-■first-■slice).

• **Docker is packaged, not proven.■** No container has been built on the

development host (no Docker/■WSL); the CI workflow and `scripts/■check_■docker.■py` are ready but unexecuted there.■ The official judge path could therefore fail on first contact; `scripts/■check_■clean_■start.■py` covers the non-■container path.■ Evidence: [PIPELINE_■VERIFICATION](docs/■PIPELINE_■VERIFICATION.■md).

• **One previously flaky test.■** `test_■mutation_■requires_■token_■and_■matching_■origin`

in `tests/■test_■review.■py` failed intermittently on the development host; it passed five consecutive runs on 16 October and the fresh-■clone suite is 131/■131.■ Named so a judge re-■running it is not surprised.■ Evidence: `python -■m unittest tests.■test_■review.■ServerTests.■test_■mutation_■requires_■token_■and_■matching_■origin`.

• **Heuristic roles and languages.■** Corpus role and level tags are heuristic

hints; 13 of 45 manifest rows are unknown-■language.■ Level-■of-■government breadth claims wait for reviewed metadata.■ Evidence: [BUILD_■STATUS](docs/■BUILD_■STATUS.■md).

• **Conservative matching has blind spots.■** Whitespace-■only recovery cannot

bridge OCR reflow, and the corrupt-■name marker does not catch all corruption.■ Some real values will read as abstentions until review — accepted as the price of never guessing.■ Evidence: `src/■skony/■verify.■py`.

The local workbench displays PDF page images, source text and highlighted spans.■ Reviewers can confirm, correct, record absence or undo.■ Decisions are separate from model suggestions in SQLite, with before/■after audit history.■ Visual transcriptions are explicitly labelled and carry no claimed text offsets.■ Exports contain reviewed fields only, plus a list of remaining unreviewed fields.■ Reviewer identity is a self-■entered label; multi-■user authentication is not implemented.

No OCR, multi-■page field assembly, controlled vocabulary mapping, cross-■body query interface or gold-■set evaluation is implemented yet; the seam named above is design work, not completed work.■ The Docker runner is packaged but has not been built/■run in this environment.■ Whitespace-■only matching recovers some layout differences but preserves wording, punctuation and original offsets.■ Other OCR/■layout differences still require review; the OCR marker heuristic does not detect all corrupt names.■ Copied text can still be assigned to the wrong semantic field.■ Model identity must be confirmed against the provider's documented served-■model mapping.

Deployment and licensing

The application uses a configurable Apertus endpoint.■ A Swiss endpoint and self-■hosted endpoint are deployment targets; air-■gapped operation has not been tested.

Component	Licence	Note
Application code (this repository)	Apache-■2.■0	per event terms
Documentation and this report	CC-■BY-■4.■0	per event terms
Bundled synthetic fixture (`data/■sample/■`)	CDLA-■Permissive-■2.■0	authored; fictional people and parliament
Runtime dependencies: pydantic, httpx, pypdf (pinned in `requirements.■txt`)	MIT /■ BSD-■3-■Clause	check each package `LICENSE` before redistribution
Apertus v1.■5 via the CSCS endpoint (development time)	Apache-■2.■0 (model)	not redistributed
OpenParlData source PDFs (private corpus)	source-■specific terms	not redistributed; a source manifest must state item-■level licences before any redistribution or before a dataset licence is applied to our own annotations
Container base (`python:3.■11-■slim`, Poppler)	Debian package terms	runner image only